Ashley Dhar

Boston, MA · (475) 419-4665 · ashdhar@bu.edu · LinkedIn · Portfolio

RESEARCH INTERESTS

Biomaterials design & characterization · Scaffold fabrication for therapeutic delivery · Tissue engineering · Biocompatibility and cell–material interactions · Bioinstrumentation

EDUCATION

Bachelor of Science in Biomedical Engineering — Expected May 2027
Boston University · GPA: 3.5/4.0 · Dean's List (2024–2025) — Boston, MA
Coursework: Biomedical Measurements, Signals & Controls, Organic Chemistry, Molecular & Cell Biology, Engineering Mechanics, Circuits

Biomedical Engineering Coursework — Aug 2023 – May 2024
University of Connecticut — Storrs, CT

RESEARCH EXPERIENCE

Undergraduate Research Assistant — Apr 2025 – Apr 2026
Dental Biomaterials Laboratory, Housman Medical Research Center, Boston University — Boston, MA

- Assisted in characterizing biomaterial scaffolds using FTIR spectroscopy to identify functional groups and support structure-property analysis for therapeutic delivery research
- Supported SEM imaging workflows to examine nanostructure morphology, porosity, and surface topography of candidate materials
- Helped run mammalian cell culture experiments, including MTT viability assays and cytotoxicity screening, to assess biocompatibility under aseptic conditions
- Contributed to a bacterial adhesion study culturing *E. coli* on dental material surfaces; helped collect and interpret data on how surface hardness and corrosion affect microbial colonization

Student Researcher — Sept 2022 – May 2023
Danbury Hospital, Nuvance Health — Danbury, CT

- Rotated across microbiology, hematology, clinical chemistry, blood bank, diagnostics, and histology divisions; observed workflows and instrumentation including centrifuges, flow cytometers, and automated analyzers in a clinical laboratory setting
- Assisted in processing histopathology specimens using tissue fixation, microtome sectioning, and H&E staining; observed diagnostic workflows alongside pathologists and developed foundational understanding of disease pathology

PROJECTS

Dynamic Light Scattering Analysis of Yogurt Nanoparticles — Spring 2026
Biomedical Measurements (BE493), Boston University — Boston, MA

- Characterized nanoparticle size distributions and polydispersity index across regular, Greek, and drinkable yogurt using a Litesizer 500; designed sample preparation protocol and dilution series
- Evaluated statistical differences across groups using one-way ANOVA with Tukey HSD post-hoc testing; authored full technical lab report on findings

Point-to-Point IR Optical Communication System — Spring 2026
BU Photonics Center / EK 210 — Boston, MA

- Built a wireless IR communication system using Arduino microcontrollers, a 950nm IR LED, and a phototransistor receiver with op-amp signal amplification; implemented pulse-width bit encoding and decoding
- Troubleshot signal instability and improved decoding accuracy from ∼30% to ∼100% by optimizing LED drive current and encoding timing

Howe-Warren Hybrid Truss Structural Analysis — Spring 2025
Engineering Mechanics (EK 301), Boston University — Boston, MA

- Designed acrylic truss to maximize load-to-cost ratio; performed buckling analysis across three failure cases; predicted failure load of 61.51 oz with member m4 as the critical compression member

OTHER EXPERIENCE

AI Data Annotator, RWS Group & Mercor (Remote) — Jan 2025 – Present
Pharmacy Associate, CVS Pharmacy (Fairfield, CT) — Dec 2023 – Sept 2024

TECHNICAL SKILLS

Laboratory: FTIR Spectroscopy, SEM, Dynamic Light Scattering (DLS), Mammalian Cell Culture, MTT Assays, Cytotoxicity Testing, Biomaterials Characterization, PCR, Microtomy, Aseptic Technique, Histological Preparation
Computational: MATLAB, Python, Arduino (C), SolidWorks, LaTeX, Microsoft Office, Google Workspace
Languages: English (Native), French (Conversational)

LEADERSHIP, ACTIVITIES & AWARDS

Member, Biomedical Engineering Society (BMES), Boston University — Sept 2024 – Present
Dean's List (BU, 2024–2025) · Exemplary Work in Science/Engineering, CT STEM Foundation (2023) · Bausch & Lomb Honorary Science Award (2022)